

МІНІСТЕРСТВО ОСВІТИ І НАУКИ УКРАЇНИ
НАЦІОНАЛЬНИЙ УНІВЕРСИТЕТ «ПОЛТАВСЬКА ПОЛІТЕХНІКА
ІМЕНІ ЮРІЯ КОНДРАТЮКА»

Навчально-науковий інститут інформаційних технологій і робототехніки
Кафедра комп'ютерних та інформаційних технологій і систем

Лабораторна робота № 3

з навчальної дисципліни

"Технології розробки корпоративних Web-додатків"

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Очікуваний результат (Звіт та код):

1. **Посилання на Git-репозиторій.**
2. **Скріншоти роботи API:**
 1. Для REST: Swagger UI або Postman.
 1. Для GraphQL: Скріншот запиту (Query) та мутації (Mutation) у Playground.
 1. Для gRPC: Скріншот виклику методу через gRPC клієнт (напр. Postman gRPC або BloomRPC).
3. **Приклади коду** (фрагменти у звіті):
 1. Визначення моделі API (DTO / Proto message / GraphQL Type).
 1. Реалізація методу (Controller Action / Resolver / gRPC Service method).

Посилання на Git-репозиторій:

https://github.com/tkvadym/TRVD_2026_402TN_Tkachenko_Labs

REST: Swagger UI:

Скріни всіх передбачених API:

Taskerito API (Lab 3) 1.0.0 OAS 3.1

/openapi.json

Users

GET	/users/	Get All Users	⌵
POST	/users/	Create User	⌵
GET	/users/{user_id}	Get User	⌵
PATCH	/users/{user_id}	Update User	⌵

Projects

POST	/projects/	Create Project	⌵
GET	/projects/{project_id}	Get Project	⌵
PATCH	/projects/{project_id}	Update Project	⌵
GET	/projects/user/{user_id}	Get User Projects	⌵

Project Members

POST	/projects/{project_id}/members/{user_id}	Add Member	⌵
DELETE	/projects/{project_id}/members/{user_id}	Remove Member	⌵
GET	/projects/{project_id}/members/	Get Project Members	⌵

Tasks



POST	/tasks/ Create Task	 ▼
GET	/tasks/{task_id} Get Task	▼
PATCH	/tasks/{task_id} Update Task	▼
DELETE	/tasks/{task_id} Delete Task	▼
PATCH	/tasks/{task_id}/status Update Task Status	▼

Comments



POST	/comments/ Add Comment	▼
GET	/comments/task/{task_id} Get Task Comments	▼

Dashboard



GET	/dashboard/{project_id} Get Project Dashboard	▼
-----	---	---

Yci schemas:

Schemas

CommentCreate ^ Collapse all object

content* string ≥ 1 characters

task_id* integer

CommentRead ^ Collapse all object

content* string ≥ 1 characters

id* integer

task_id* integer

user_id* integer

created_at* string date-time

DashboardStats ^ Collapse all object

total_tasks* integer

tasks_to_do* integer

tasks_in_progress* integer

tasks_done* integer

overdue_tasks_count* integer

HTTPValidationError ^ Collapse all object

detail > Expand all array<object>

ProjectCreate ^ Collapse all object

name* string [3, 200] characters

description > Expand all (string | null)

status > Expand all string [1, 50] characters

ProjectRead ^ Collapse all object

name* string [3, 200] characters

description > Expand all (string | null)

status > Expand all string [1, 50] characters

id* integer

created_by* integer

created_at* string date-time

ProjectUpdate ^ Collapse all object

name > Expand all (string | null)
description > Expand all (string | null)
status > Expand all (string | null)

TaskCreate ^ Collapse all object

title* string [3, 250] characters
description > Expand all (string | null)
status > Expand all string [1, 50] characters
priority > Expand all string [1, 50] characters
project_id* integer
assigned_to > Expand all (integer | null)
deadline > Expand all (string | null)

TaskRead ^ Collapse all object

title* string [3, 250] characters
description > Expand all (string | null)
status > Expand all string [1, 50] characters
priority > Expand all string [1, 50] characters
id* integer
project_id* integer
created_by* integer
assigned_to* > Expand all (integer | null)
deadline* > Expand all (string | null)
created_at* string date-time
updated_at* string date-time

TaskStatusUpdate ^ Collapse all object

status* string [1, 50] characters

TaskUpdate ^ Collapse all object

title > Expand all (string | null)
description > Expand all (string | null)
priority > Expand all (string | null)
assigned_to > Expand all (integer | null)
deadline > Expand all (string | null)

UserCreate ^ [Collapse all](#) object

```
full_name* string [2, 150] characters
email* string email
password* > Expand all string ≥ 8 characters
role_id* integer
```

UserRead ^ [Collapse all](#) object

```
full_name* string [2, 150] characters
email* string email
id* integer
role_id* integer
is_active* boolean
created_at* string date-time
```

UserUpdate ^ [Collapse all](#) object

```
full_name > Expand all (string | null)
role_id > Expand all (integer | null)
is_active > Expand all (boolean | null)
```

ValidationError ^ [Collapse all](#) object

```
loc* > Expand all array<(string | integer)>
msg* string
type* string
input any
ctx object
```

Демонстрація роботи endpoints:

POST

/users/ Create User

Parameters

No parameters

Request body required

application/json

Edit Value | Schema

```
{  "full_name": "Vadym Tkachenko",  "email": "tkachenko@gmail.com",  "password": "12345rethghxt678",  "role_id": 0}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \  'http://127.0.0.1:8000/users/' \  -H 'accept: application/json' \  -H 'Content-Type: application/json' \  -d '{  "full_name": "Vadym Tkachenko",  "email": "tkachenko@gmail.com",  "password": "12345rethghxt678",  "role_id": 0}'
```

Request URL

http://127.0.0.1:8000/users/

Server response

Code	Details
201	Response body <pre>{ "full_name": "Vadym Tkachenko", "email": "tkachenko@gmail.com", "id": 1, "role_id": 0, "is_active": true, "created_at": "2026-03-27T11:58:37.331287"}</pre> Download

GET

/users/{user_id} Get User

Parameters

Name	Description
user_id * required	
integer	
(path)	

1

Execute

Clear

Responses

Curl

```
curl -X 'GET' \  'http://127.0.0.1:8000/users/1' \  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/users/1

Server response

Code	Details
200	Response body <pre>{ "full_name": "Vadym Tkachenko", "email": "tkachenko@gmail.com", "id": 1, "role_id": 0, "is_active": true, "created_at": "2026-03-27T11:58:37.331287"}</pre> Download Response headers <pre>content-length: 139 content-type: application/json date: Fri,27 Mar 2026 12:00:02 GMT server: uvicorn</pre>

Users

GET

/users/ Get All Users

Cancel

No parameters

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/users/' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/users/
```

Server response

Code

Details

200

Response body

```
{
  {
    "full_name": "Vadym Tkachenko",
    "email": "tkachenko@gmail.com",
    "id": 1,
    "role_id": 0,
    "is_active": true,
    "created_at": "2026-03-27T11:58:37.331287"
  }
}
```

Download

Response headers

```
content-length: 141
content-type: application/json
date: Fri, 27 Mar 2026 12:00:24 GMT
server: uvicorn
```

Responses

PATCH

/users/{user_id} Update User

CancelReset

Name

Description

user_id *

required

integer

(path)

1

Request body

required

application/json

Edit Value | Schema

```
{
  "role_id": 1
}
```

ExecuteClear

Responses

Curl

```
curl -X 'PATCH' \
  'http://127.0.0.1:8000/users/1' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "role_id": 1
  }'
```

Request URL

```
http://127.0.0.1:8000/users/1
```

Server response

Server response

Code

Details

200

Response body

```
{
  "full_name": "Vadym Tkachenko",
  "email": "tkachenko@gmail.com",
  "id": 1,
  "role_id": 1,
  "is_active": true,
  "created_at": "2026-03-27T11:58:37.331287"
}
```

Download

Response headers

```
content-length: 139
content-type: application/json
date: Fri, 27 Mar 2026 12:02:09 GMT
server: uvicorn
```

Responses

POST

/projects/ Create Project

Cancel

Reset

Parameters

Name	Description
current_user_id * required integer (query)	ID of the user creating project 1

Request body required

application/json

Edit Value | Schema

```
{
  "name": "Random name",
  "description": "Random desc",
  "status": "Active"
}
```

Execute

Clear

Responses

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/projects/?current_user_id=1' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Random name",
    "description": "Random desc",
    "status": "Active"
  }'
```

Request URL

http://127.0.0.1:8000/projects/?current_user_id=1

Server response

Code

Details

201

Response body

```
{
  "name": "Random name",
  "description": "Random desc",
  "status": "Active",
  "id": 1,
  "created_by": 1,
  "created_at": "2026-03-27T12:03:35.692198"
}
```

Download

Response headers

```
content-length: 132
content-type: application/json
date: Fri, 27 Mar 2026 12:03:35 GMT
server: uvicorn
```

Responses

Code	Description	Links
------	-------------	-------

GET

/projects/{project_id} Get Project

Cancel

Parameters

Name	Description
project_id * required	
integer	1
(path)	

ExecuteClear

Responses

Curl

curl -X 'GET' \n'http://127.0.0.1:8000/projects/1' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:8000/projects/1

Server response

Code	Details
200	Response body {\n "name": "Random name",\n "description": "Random desc",\n "status": "Active",\n "id": 1,\n "created_by": 1,\n "created_at": "2026-03-27T12:03:35.692198"\n} Download Response headers content-length: 132\ncontent-type: application/json\ndate: Fri, 27 Mar 2026 12:04:29 GMT\nserver: uvicorn

Responses

GET

/projects/user/{user_id} Get User Projects

Cancel

Parameters

Name	Description
user_id * required	
integer	0
(path)	

ExecuteClear

Responses

Curl

curl -X 'GET' \n'http://127.0.0.1:8000/projects/user/0' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:8000/projects/user/0

Server response

Code	Details
200	Response body [] Download Response headers content-length: 2\ncontent-type: application/json\ndate: Fri, 27 Mar 2026 12:04:57 GMT\nserver: uvicorn

Responses

GET

/projects/user/{user_id} Get User Projects

Parameters

Cancel

Name	Description
user_id required integer (path)	1

Execute

Clear

Responses

Curl

curl -X 'GET' \
'http://127.0.0.1:8000/projects/user/1' \
-H 'accept: application/json'

Request URL

http://127.0.0.1:8000/projects/user/1

Server response

Code	Details
200	Response body <pre>{ "name": "Random name", "description": "Random desc", "status": "Active", "id": 1, "created_by": 1, "created_at": "2026-03-27T12:03:35.692198" }</pre> Download Response headers content-length: 134 content-type: application/json date: Fri,27 Mar 2026 12:05:03 GMT server: uvicorn

Responses

PATCH

/projects/{project_id} Update Project

Parameters

Cancel

Reset

Name	Description
project_id required integer (path)	1

Request body required

application/json

Edit Value | Schema

```
{  
  "description": "Another random desc"  
}
```

Execute

Clear

Responses

Curl

```
curl -X 'PATCH' \
  'http://127.0.0.1:8000/projects/1' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "description": "Another random desc"
  }'
```

Request URL

http://127.0.0.1:8000/projects/1

Server response

Code Details

200

Response body

```
{
  "name": "Random name",
  "description": "Another random desc",
  "status": "Active",
  "id": 1,
  "created_by": 1,
  "created_at": "2026-03-27T12:03:35.692198"
}
```

Response headers

```
content-length: 140
content-type: application/json
date: Fri, 27 Mar 2026 12:05:58 GMT
server: uvicorn
```

Responses

Code Description

Link

POST /projects/{project_id}/members/{user_id} Add Member

Parameters

Cancel

Name Description

project_id * required
integer
(path)

1

user_id * required
integer
(path)

1

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/projects/1/members/1' \
  -H 'accept: application/json' \
  -d ''
```

Request URL

http://127.0.0.1:8000/projects/1/members/1

Server response

Code Details

201

Response body

```
{
  "message": "Member added successfully"
}
```

Response headers

```
content-length: 39
content-type: application/json
date: Fri, 27 Mar 2026 12:07:12 GMT
server: uvicorn
```

Responses

GET

/projects/{project_id}/members/ Get Project Members

Cancel

Parameters

Name	Description
project_id * required	
integer	1
(path)	

ExecuteClear

Responses

Curl

curl -X 'GET' \ 'http://127.0.0.1:8000/projects/1/members/' \ -H 'accept: application/json'

Request URL

http://127.0.0.1:8000/projects/1/members/

Server response

Code	Details
200	Response body [{"user_id": 1, "joined_at": "2026-03-27T12:07:12.640636"}] Response headers content-length: 56 content-type: application/json date: Fri, 27 Mar 2026 12:07:26 GMT server: uvicorn Download

DELETE

/projects/{project_id}/members/{user_id} Remove Member

Cancel

Parameters

Name	Description
project_id * required	
integer	1
(path)	
user_id * required	
integer	1
(path)	

ExecuteClear

Responses

Curl

curl -X 'DELETE' \ 'http://127.0.0.1:8000/projects/1/members/1' \ -H 'accept: */*'

Request URL

http://127.0.0.1:8000/projects/1/members/1

Server response

Code	Details
204	Response headers content-type: application/json date: Fri, 27 Mar 2026 12:08:14 GMT server: uvicorn

Tasks

POST

/tasks/ Create Task

Parameters

Cancel

Reset

Name

Description

current_user_id

required

ID of the task author

integer

(query)

1

Request body

required

application/json

Edit Value | Schema

{
 "title": "Random task title",
 "description": "Random task desc",
 "status": "To Do",
 "priority": "Medium",
 "project_id": 1,
 "assigned_to": 1,
 "deadline": "2026-03-27T12:09:09.087Z"
}

Execute

Clear

Responses

Curl

```
curl -X 'POST' \  
  'https://127.0.0.1:8000/tasks/?current_user_id=1' \  
  -H 'accept: application/json' \  
  -H 'Content-Type: application/json' \  
  -d '{  
    "title": "Random task title",  
    "description": "Random task desc",  
    "status": "To Do",  
    "priority": "Medium",  
    "project_id": 1,  
    "assigned_to": 1,  
    "deadline": "2026-03-27T12:09:09.087Z"  
  }'  
'
```

Request URL

http://127.0.0.1:8000/tasks/?current_user_id=1

Request URL

http://127.0.0.1:8000/tasks/?current_user_id=1

Server response

Code

Details

201

Response body

{
 "title": "Random task title",
 "description": "Random task desc",
 "status": "To Do",
 "priority": "Medium",
 "id": 1,
 "project_id": 1,
 "created_by": 1,
 "assigned_to": 1,
 "deadline": "2026-03-27T12:09:09.087000",
 "created_at": "2026-03-27T12:10:04.623421",
 "updated_at": "2026-03-27T12:10:04.623421"
}

Download

Response headers

content-length: 276
content-type: application/json
date: Fri, 27 Mar 2026 12:10:03 GMT
server: uvicorn

Responses

GET

/tasks/{task_id} Get Task

⌵

Parameters

Cancel

Name	Description
task_id * required	
integer	
(path)	
1	

ExecuteClear

Responses

Curl

curl -X 'GET' \n'http://127.0.0.1:8080/tasks/1' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:8080/tasks/1

Server response

Code

Details

200

Response body

{\n "title": "Random task title",\n "description": "Random task desc",\n "status": "To Do",\n "priority": "Medium",\n "id": 1,\n "project_id": 1,\n "created_by": 1,\n "assigned_to": 1,\n "deadline": "2026-03-27T12:09:09.007000",\n "created_at": "2026-03-27T12:10:04.623421",\n "updated_at": "2026-03-27T12:10:04.623421"\n}

Download

Response headers

Content-Length: 376

PATCH

/tasks/{task_id}/status Update Task Status

⌵

Parameters

Cancel

Reset

Name	Description
task_id * required	
integer	
(path)	
1	

Request body

required

application/json

Edit Value

Schema

{\n "status": "Completed"\n}

Execute

Clear

Responses

Curl

```
curl -X 'PATCH' \
  'http://127.0.0.1:8000/tasks/1/status' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "status": "Completed"
  }'
```

Request URL

http://127.0.0.1:8000/tasks/1/status

Server response

Code

Details

200

Response body

```
{
  "title": "Random task title",
  "description": "Random task desc",
  "status": "Completed",
  "priority": "Medium",
  "id": 1,
  "project_id": 1,
  "created_by": 1,
  "assigned_to": 1,
  "deadline": "2026-03-27T12:09:09.087000",
  "created_at": "2026-03-27T12:10:04.623421",
  "updated_at": "2026-03-27T12:11:14.591938"
}
```

Response headers

```
content-length: 280
content-type: application/json
date: Fri, 27 Mar 2026 12:11:14 GMT
server: uvicorn
```

Responses

DELETE /tasks/{task_id} Delete Task

Parameters

Cancel

Name	Description
task_id	required integer (path)
	1

Execute

Clear

Responses

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/tasks/1' \
  -H 'accept: */*'
```

Request URL

http://127.0.0.1:8000/tasks/1

Server response

Code

Details

204

Response headers

```
content-type: application/json
date: Fri, 27 Mar 2026 12:12:00 GMT
server: uvicorn
```

Responses

Code	Description	Links
204	Successful Response	No links

Comments

POST /comments/ Add Comment

Parameters

Cancel

Reset

Name

Description

current_user_id * required

integer (query)

1

Request body required

application/json

Edit Value | Schema

{
 "content": "Random comment",
 "task_id": 1
}

Execute

Clear

Responses

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/comments/?current_user_id=1' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{  
  "content": "Random comment",  
  "task_id": 1  
}'
```

Request URL

```
http://127.0.0.1:8000/comments/?current_user_id=1
```

Server response

Code

Details

201

Response body

```
{  
  "content": "Random comment",  
  "id": 1,  
  "task_id": 1,  
  "user_id": 1,  
  "created_at": "2026-03-27T12:12:39.925985"  
}
```

Download

Response headers

```
content-length: 101  
content-type: application/json  
date: Fri, 27 Mar 2026 12:12:39 GMT  
server: uvicorn
```

Responses

Code

Description

Links

GET

/comments/task/{task_id}

Get Task Comments

^

Parameters

Cancel

Name	Description
task_id required	
integer	
(path)	

ExecuteClear

Responses

Curl

curl -X 'GET' \n'http://127.0.0.1:8000/comments/task/1' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:8000/comments/task/1

Server response

Code

Details

200

Response body

[
 {
 "content": "Random comment",
 "id": 1,
 "task_id": 1,
 "user_id": 1,
 "created_at": "2026-03-27T12:12:39.925985"
 }
]

Download

Response headers

content-length: 103
content-type: application/json
date: Fri, 27 Mar 2026 12:13:03 GMT
server: uvicorn

Responses

Name

Description

Link

Dashboard

GET

/dashboard/{project_id}

Get Project Dashboard

^

Parameters

Cancel

Name	Description
project_id required	
integer	
(path)	

ExecuteClear

Responses

Curl

curl -X 'GET' \n'http://127.0.0.1:8000/dashboard/1' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:8000/dashboard/1

Server response

Code

Details

200

Response body

{
 "total_tasks": 1,
 "tasks_to_do": 1,
 "tasks_in_progress": 0,
 "tasks_done": 0,
 "overdue_tasks_count": 1
}

Download

Response headers

content-length: 94
content-type: application/json
date: Fri, 27 Mar 2026 12:13:20 GMT
server: uvicorn

Responses

Приклади коду:

```
task.py x
schemas > task.py > BaseModel
1 from pydantic import BaseModel, Field, ConfigDict
2 from typing import Optional
3 from datetime import datetime
4
5 class TaskBase(BaseModel):
6     title: str = Field(..., min_length=3, max_length=250)
7     description: Optional[str] = None
8     status: str = Field("To Do", min_length=1, max_length=50)
9     priority: str = Field("Medium", min_length=1, max_length=50)
10
11 class TaskCreate(TaskBase):
12     project_id: int
13     assigned_to: Optional[int] = None
14     deadline: Optional[datetime] = None
15
16 class TaskUpdate(BaseModel):
17     title: Optional[str] = Field(None, min_length=3, max_length=250)
18     description: Optional[str] = None
19     priority: Optional[str] = Field(None, min_length=1, max_length=50)
20     assigned_to: Optional[int] = None
21     deadline: Optional[datetime] = None
22
23 class TaskStatusUpdate(BaseModel):
24     status: str = Field(..., min_length=1, max_length=50)
25
26 class TaskRead(TaskBase):
27     id: int
28     project_id: int
29     created_by: int
30     assigned_to: Optional[int]
31     deadline: Optional[datetime]
32     created_at: datetime
33     updated_at: datetime
34
35     model_config = ConfigDict(from_attributes=True)
36
```

```
1 from sqlalchemy.orm import Session
2 from models.task import Task
3 from schemas.task import TaskCreate, TaskUpdate, TaskStatusUpdate
4 from repositories.task_repository import TaskRepository
5 from repositories.project_repository import ProjectRepository
6
7 class TaskService:
8     def __init__(self, db: Session):
9         self.repo = TaskRepository(db)
10        self.project_repo = ProjectRepository(db)
11
12    def create_task(self, task_create: TaskCreate, current_user_id: int) -> Task:
13        if not self.project_repo.get_by_id(task_create.project_id):
14            raise ValueError(f"Project {task_create.project_id} not found")
15
16        db_task = Task(
17            project_id=task_create.project_id,
18            title=task_create.title,
19            description=task_create.description,
20            status=task_create.status,
21            priority=task_create.priority,
22            assigned_to=task_create.assigned_to,
23            created_by=current_user_id,
24            deadline=task_create.deadline
25        )
26        return self.repo.create(db_task)
27
28    def get_task(self, task_id: int) -> Task:
29        task = self.repo.get_by_id(task_id)
30        if not task:
31            raise ValueError(f"Task {task_id} not found")
32        return task
33
34    def update_task(self, task_id: int, task_update: TaskUpdate) -> Task:
35        task = self.get_task(task_id)
36
37        if task_update.title is not None:
38            task.title = task_update.title
39        if task_update.description is not None:
40            task.description = task_update.description
41        if task_update.priority is not None:
42            task.priority = task_update.priority
43        if task_update.assigned_to is not None:
44            task.assigned_to = task_update.assigned_to
45        if task_update.deadline is not None:
46            task.deadline = task_update.deadline
47
48        return self.repo.update(task)
49
50    def update_task_status(self, task_id: int, status_update: TaskStatusUpdate) -> Task:
51        task = self.get_task(task_id)
52        task.status = status_update.status
53        return self.repo.update(task)
54
55    def delete_task(self, task_id: int):
56        task = self.get_task(task_id)
57        self.repo.delete(task)
58
```

```
1  from sqlalchemy.orm import Session
2  from models.task import Task
3  from datetime import datetime
4
5  class TaskRepository:
6      def __init__(self, db: Session):
7          self.db = db
8
9      def get_by_id(self, task_id: int):
10         return self.db.query(Task).filter(Task.id == task_id).first()
11
12     def get_by_project(self, project_id: int):
13         return self.db.query(Task).filter(Task.project_id == project_id).all()
14
15     def get_by_assignee(self, user_id: int):
16         return self.db.query(Task).filter(Task.assigned_to == user_id).all()
17
18     def get_overdue(self, project_id: int):
19         now = datetime.utcnow()
20         return self.db.query(Task).filter(
21             Task.project_id == project_id,
22             Task.deadline < now,
23             Task.status != 'Done',
24             Task.status != 'Cancelled'
25         ).all()
26
27     def create(self, task: Task):
28         self.db.add(task)
29         self.db.commit()
30         self.db.refresh(task)
31         return task
32
33     def update(self, task: Task):
34         self.db.commit()
35         self.db.refresh(task)
36         return task
37
38     def delete(self, task: Task):
39         self.db.delete(task)
40         self.db.commit()
41
```

```

task.py X
models > task.py > ...
1 from sqlalchemy import Column, Integer, String, Text, DateTime, ForeignKey
2 from sqlalchemy.orm import relationship
3 from datetime import datetime
4 from .base import Base
5
6 class Task(Base):
7     __tablename__ = "tasks"
8
9     id = Column(Integer, primary_key=True, autoincrement=True)
10    project_id = Column(Integer, ForeignKey("projects.id"), nullable=False)
11    title = Column(String(250), nullable=False)
12    description = Column(Text, nullable=True)
13    status = Column(String(50), nullable=False, default='To Do')
14    priority = Column(String(50), nullable=False, default='Medium')
15    assigned_to = Column(Integer, ForeignKey("users.id"), nullable=True)
16    created_by = Column(Integer, ForeignKey("users.id"), nullable=False)
17    deadline = Column(DateTime, nullable=True)
18    created_at = Column(DateTime, nullable=False, default=datetime.utcnow)
19    updated_at = Column(DateTime, nullable=False, default=datetime.utcnow, onupdate=datetime.utcnow)
20
21    project = relationship("Project", backref="tasks")
22    assignee = relationship("User", foreign_keys=[assigned_to])
23    author = relationship("User", foreign_keys=[created_by])
24    Ctrl+L for Agent

```

```

tasks_router.py X
routes > tasks_router.py > APIRouter
1 from fastapi import APIRouter, Depends, status, Query
2 from sqlalchemy.orm import Session
3 from typing import List
4
5 from schemas.task import TaskCreate, TaskRead, TaskUpdate, TaskStatusUpdate
6 from services.task_service import TaskService
7 from dependencies import get_db
8
9 router = APIRouter(prefix="/tasks", tags=["Tasks"])
10
11 @router.post("/", response_model=TaskRead, status_code=status.HTTP_201_CREATED)
12 def create_task(task: TaskCreate, current_user_id: int = Query(..., description="ID of the task author"), db: Session = Depends(get_db)):
13     service = TaskService(db)
14     return service.create_task(task, current_user_id)
15
16 @router.get("/{task_id}", response_model=TaskRead)
17 def get_task(task_id: int, db: Session = Depends(get_db)):
18     service = TaskService(db)
19     return service.get_task(task_id)
20
21 @router.patch("/{task_id}", response_model=TaskRead)
22 def update_task(task_id: int, task_update: TaskUpdate, db: Session = Depends(get_db)):
23     service = TaskService(db)
24     return service.update_task(task_id, task_update)
25
26 @router.patch("/{task_id}/status", response_model=TaskRead)
27 def update_task_status(task_id: int, status_update: TaskStatusUpdate, db: Session = Depends(get_db)):
28     service = TaskService(db)
29     return service.update_task_status(task_id, status_update)
30
31 @router.delete("/{task_id}", status_code=status.HTTP_204_NO_CONTENT)
32 def delete_task(task_id: int, db: Session = Depends(get_db)):
33     service = TaskService(db)
34     service.delete_task(task_id)
35     return None
36

```